Gas pipeline KP 0-8.5 corridor survey

Drone mission field brief · DJI Matrice 350 RTK + P1 · flight date Fri 16 Oct 2026 · prepared 13 Oct 2026

Summary

Flight date **Fri 16 Oct 2026 (UTC+8)**	Aircraft **DJI Matrice 350 RTK + P1**	Mission **route**	Area **59.39 ha**
Route **8.48 km, 4 sections**	Height **88 m**	Return-to-home height **88 m**	Battery-change height **88 m**
GSD **1.10 cm/px**	Speed **5.5 m/s**	Photos **1866**	Image coverage **117.29 ha**
Flights **4**	Flight time **1h 51m**	Usable / battery **33.8 min**	Wind **10 kn (5.0 m/s) from 250°**
Temp / payload **24 °C / 0.00 kg**	Accessory power **0 W**	Daylight **05:36 – 18:29**	Civil twilight **05:11 – 18:55**
Best light (sun >= 30°) **08:03 – 16:03**	On site **07:00 – 10:00 (3.0 h)**	Battery sets **6 (recommended 6)**	Charge waiting **0 min**

Mission map

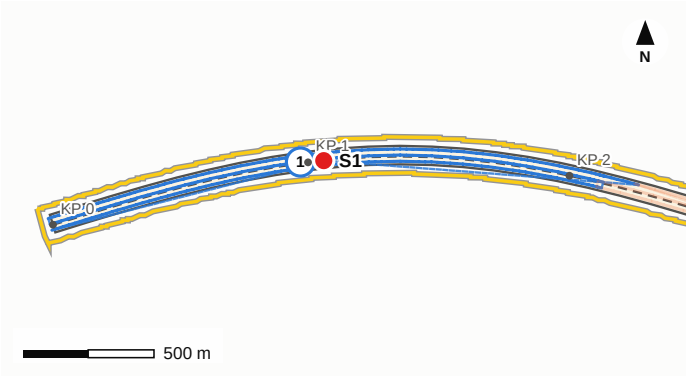


Flight 1 · Flight 2 · Flight 3 · Flight 4 · Transit to / from launch station · Take-off / launch station · Image coverage (footprint extent)

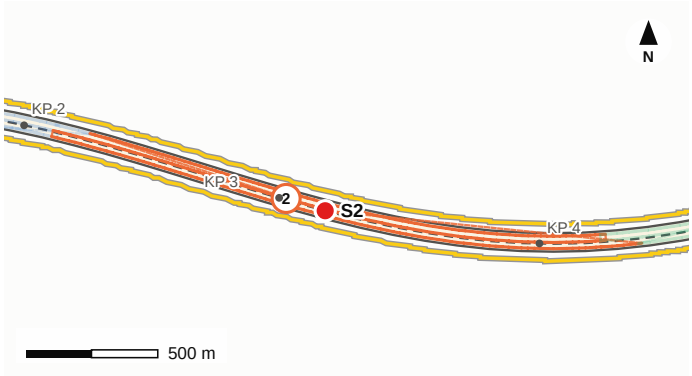
Grid north. Map and plan in the mission's planning coordinate system.

Section maps

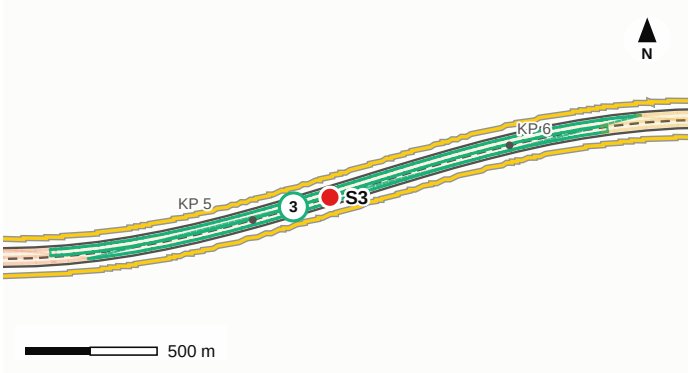
S1 · flight 1 · KP 0.00–2.26



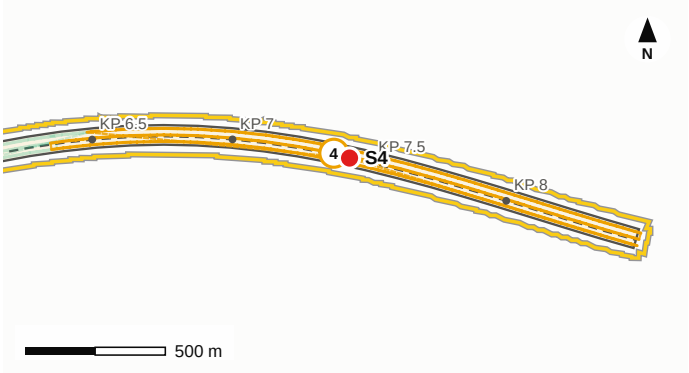
S2 · flight 2 · KP 2.11–4.38



S3 · flight 3 · KP 4.23–6.50



S4 · flight 4 · KP 6.35–8.48



Launch stations

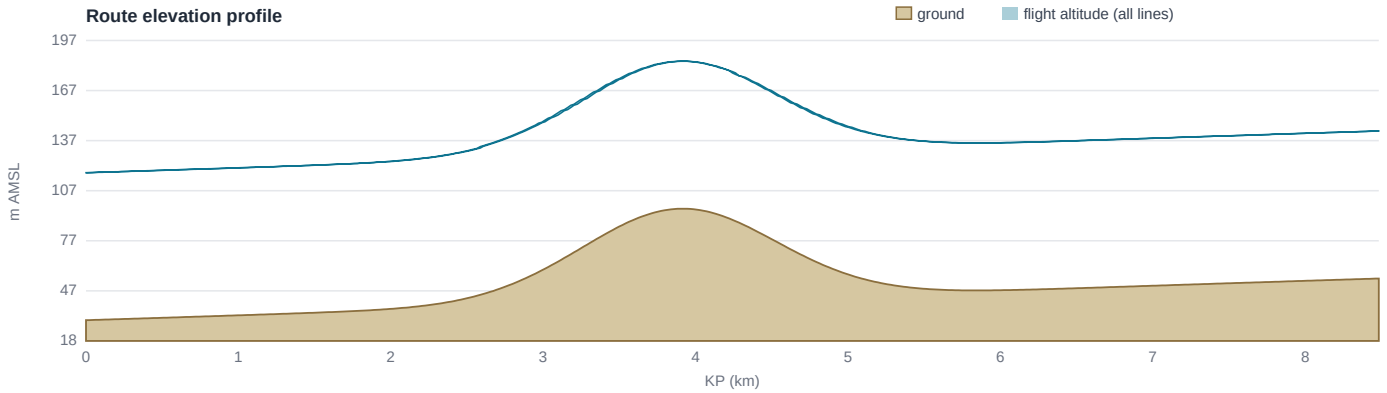
Station	Lat, Lon (WGS84)	Flights launched
S1	-32.276346, 115.747320	1
S2	-32.279983, 115.769243	1
S3	-32.279244, 115.791366	1
S4	-32.277540, 115.813465	1

Flights & battery plan

#	Launch -> land	Set	Start	End	Min	Battery	Photos	KP (km)	Sun	Notes	Done
1	S1 -> S1	1	07:15	07:43	28.5	96%	480	0.00–2.26	20–26° low sun	battery change mid-line.	☐
2	S2 -> S2	2	08:00	08:27	27.8	91%	466	2.11–4.38	29–35° low sun	battery change mid-line. drive 12 min	☐
3	S3 -> S3	3	08:44	09:12	27.9	96%	467	4.23–6.50	38–44°	battery change mid-line. drive 12 min	☐
4	S4 -> S4	4	09:28	09:55	27.0	88%	453	6.35–8.48	47–52°	drive 12 min	☐

Terrain (DEM)

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Terrain & obstacle check:
  Lowest clearance  : 88 m  (limit 30 m)
  Highest above gnd : 91 m  (limit 120 m)
  GSD over terrain  : 1.10 - 1.10 cm/px (mean 1.10)
  Height over survey: 88 m - 88 m above ground
  Worst overlap     : 75% front / 74% side (where ground is highest)
  OK - no clearance or height-limit problems found.
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Warnings & checks

- LIGHT: flights 1, 2 with the sun below 30° - long shadows reduce mapping quality. Best light 08:03–16:03 (sun >= 30°).
- Check airspace, NOTAMs, weather and the mission preview in the flight app before each flight.